

lab04_Antunes

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Lab 04

1. Carregue os pacotes readxl e tidyverse.

```
library(readxl)
library(tidyverse)
```

```
-- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
v dplyr      1.2.1      v readr      2.2.0
v forcats    1.0.1      v stringr    1.6.0
v ggplot2    4.0.3      v tibble     3.3.1
v lubridate  1.9.5      v tidyr      1.3.2
v purrr      1.2.2
-- Conflicts ----- tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()     masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(knitr)
```

2. Crie uma variável chamada fname que contenha o caminho para o arquivo WDIEXCEL.xlsx, disponível no seu diretório de dados. Confirme que o R identifica a existência do arquivo utilizando o comando file.exists

```
fname <- "datasets/WDIEXCEL.xlsx"
file.exists(fname)
```

```
[1] TRUE
```

3. Leia a planilha “Country” e armazene no objeto countries.

```
countries <- read_excel(fname, sheet = "Country")
head(countries) %>% kable(format = "latex")
```

Country Code	Short Name	Table Name	Long Name	2-alpha code
ABW	Aruba	Aruba	Aruba	AW
AFE	Africa Eastern and Southern	Africa Eastern and Southern	Africa Eastern and Southern	ZH
AFG	Afghanistan	Afghanistan	Islamic State of Afghanistan	AF
AFW	Africa Western and Central	Africa Western and Central	Africa Western and Central	ZI
AGO	Angola	Angola	People's Republic of Angola	AO
ALB	Albania	Albania	Republic of Albania	AL

4. Mantenha apenas os registros de 5 países (Brasil e outros 4 de sua escolha).

```
countries_selected <- c("BRA", "CHN", "ARG", "JPN", "NLD")

countries_filtered <- countries %>% filter(`Country Code` %in% countries_selected)

head(countries_filtered) %>% kable(format = "latex")
```

Country Code	Short Name	Table Name	Long Name	2-alpha code	Currency Unit	Special Notes
ARG	Argentina	Argentina	Argentine Republic	AR	Argentine peso	Due to the lin
BRA	Brazil	Brazil	Federative Republic of Brazil	BR	Brazilian real	NA
CHN	China	China	People's Republic of China	CN	Chinese yuan	On 1 July 199
JPN	Japan	Japan	Japan	JP	Japanese yen	Fiscal year en
NLD	Netherlands	Netherlands	Kingdom of the Netherlands	NL	Euro	A simple mult

5. Leia a planilha “Data” e armazene no objeto WDI.

```
WDI <- read_excel(fname, sheet = "Data")
head(WDI) %>% kable(format = "latex")
```

Country Name	Country Code	Indicator Name
Africa Eastern and Southern	AFE	Access to clean fuels and technologies for cooking (% of population)
Africa Eastern and Southern	AFE	Access to clean fuels and technologies for cooking, rural (% of rural population)
Africa Eastern and Southern	AFE	Access to clean fuels and technologies for cooking, urban (% of urban population)
Africa Eastern and Southern	AFE	Access to electricity (% of population)
Africa Eastern and Southern	AFE	Access to electricity, rural (% of rural population)
Africa Eastern and Southern	AFE	Access to electricity, urban (% of urban population)

6. Selecione apenas as colunas:

- Country Code
- Indicator Code
- E todos os anos de 1960-2018

```
WDI_selected <- WDI %>% select(`Country Code`, `Indicator Code`, `1960`:`2018`)
head(WDI_selected) %>% kable(format = "latex")
```

Country Code	Indicator Code	1960	1961	1962	1963	1964	1965	1966	1967	1968	1969	1970	1971
AFE	EG.CFT.ACCS.ZS	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
AFE	EG.CFT.ACCS.RU.ZS	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
AFE	EG.CFT.ACCS.UR.ZS	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
AFE	EG.ELC.ACCS.ZS	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
AFE	EG.ELC.ACCS.RU.ZS	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
AFE	EG.ELC.ACCS.UR.ZS	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA

7. Confirme se os dados wmi WDI estão em formato tidy. Se não estiverem, transforme-os para tal formato. Faça de forma que exista uma coluna referente ao ano chamada YEAR.

```
WDI_tidy <- WDI_selected %>% pivot_longer(cols = `1960`:`2018`, names_to = "YEAR", values_to = "value")
head(WDI_tidy) %>% kable(format = "latex")
```

Country Code	Indicator Code	YEAR	value
AFE	EG.CFT.ACCS.ZS	1960	NA
AFE	EG.CFT.ACCS.ZS	1961	NA
AFE	EG.CFT.ACCS.ZS	1962	NA
AFE	EG.CFT.ACCS.ZS	1963	NA
AFE	EG.CFT.ACCS.ZS	1964	NA
AFE	EG.CFT.ACCS.ZS	1965	NA

8. Confirme se a coluna referente ao ano está em formato inteiro.

```
WDI_tidy$YEAR %>% is.integer()
```

```
[1] FALSE
```

```
WDI_tidy <- WDI_tidy %>% mutate(YEAR = as.integer(YEAR))
```

```
WDI_tidy$YEAR %>% is.integer()
```

```
[1] TRUE
```

9. Para o objeto WDI, mantenha apenas os registros dos países listados em countries. Selecione as colunas:

- Country Code
- Short Name
- YEAR
- SP.DYN.CBRT.IN

```
WDI_final <- WDI_tidy %>%
  filter(`Indicator Code` == "SP.DYN.CBRT.IN") %>%
  inner_join(
    countries_filtered %>% select(`Country Code`, `Short Name`),
    by = "Country Code"
  ) %>%
  pivot_wider(
    names_from = `Indicator Code`,
    values_from = value
  ) %>%
  select(`Country Code`, `Short Name`, YEAR, SP.DYN.CBRT.IN)

head(WDI_final) %>% kable(format = "latex")
```

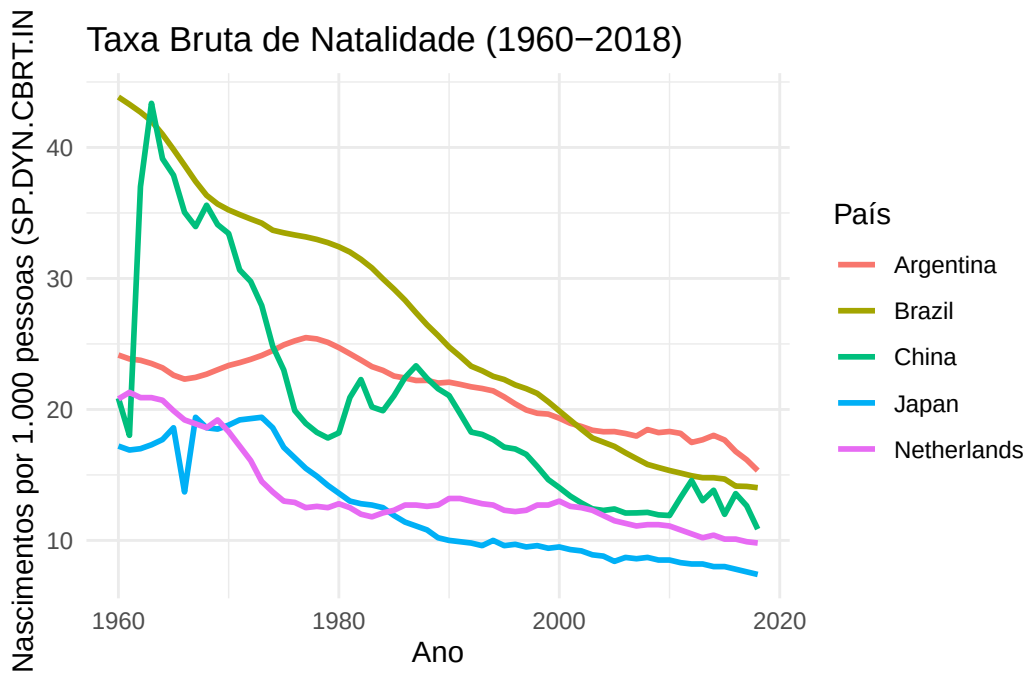
Country Code	Short Name	YEAR	SP.DYN.CBRT.IN
ARG	Argentina	1960	24.152
ARG	Argentina	1961	23.857
ARG	Argentina	1962	23.753
ARG	Argentina	1963	23.504
ARG	Argentina	1964	23.174
ARG	Argentina	1965	22.612

10. Ordene o objeto de forma que, para cada país, as informações estejam ordenadas por ano.

```
WDI_final <- WDI_final %>% arrange(`Country Code`, YEAR)
```

11. Construa um gráfico de linhas, usando o pacote ggplot2, que tenha no eixo X a variável YEAR, no eixo Y a variável SP.DYN.CBRT.IN e que cada linha corresponda a um país diferente.

```
library(ggplot2)
WDI_final %>% ggplot(aes(x = YEAR, y = SP.DYN.CBRT.IN, color = `Short Name`)) +
  geom_line(linewidth = 1) +
  labs(
    title = "Taxa Bruta de Natalidade (1960-2018)",
    x = "Ano",
    y = "Nascimentos por 1.000 pessoas (SP.DYN.CBRT.IN)",
    color = "País"
  ) +
  theme_minimal()
```



```
Sys.setenv(TZ = "America/Sao_Paulo")
Sys.time()
```

```
[1] "2026-09-07 12:15:40 -03"
```